

Implementation of NOT, AND and OR Gates Using Only NAND Gates

Boolean Expressions • Logic Diagrams • Verilog HDL • Simulation Output

1.Aim :

Implement NOT, AND, OR gates using only NAND gates in HDL. Draw the logic diagram for each and write the Boolean expression. Run the provided test scripts and capture the simulation output.

2. Logic Design & Boolean Expressions

A. NOT Gate using NAND Gate

A NOT gate inverts a single input signal. Tying both inputs of a NAND gate together forces it to operate on a single variable

// HDL (e.g., Hack HDL / Nand2Octet style)

```
CHIP Not {
```

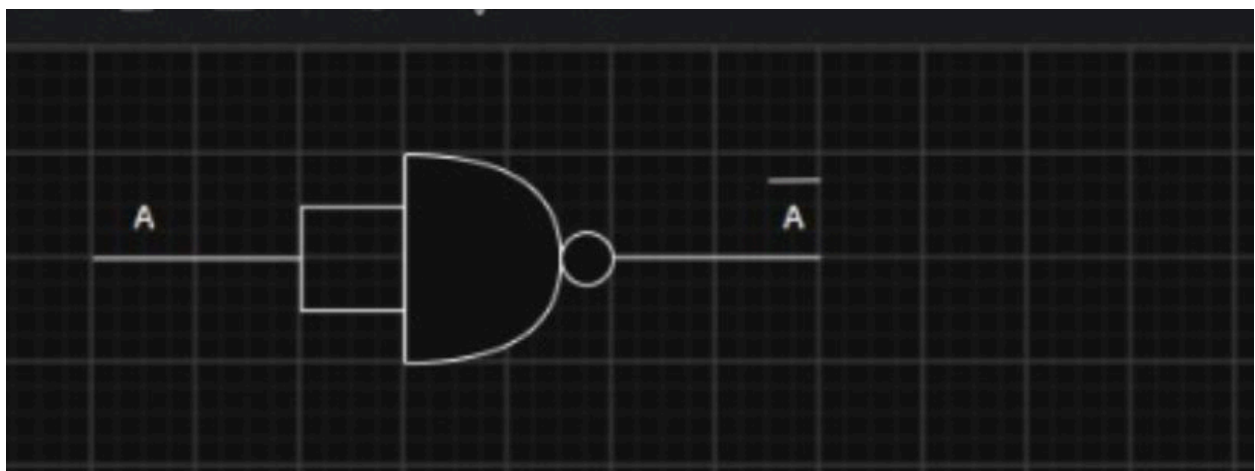
```
    IN in;
```

```
    OUT out;
```

```
    PARTS:
```

```
    Nand(a=in, b=in, out=out);
```

```
}
```



[HDL Concepts](#)[HDL Syntax](#)[Interface](#)[Simulator](#)[Test Scripts](#)[Quiz](#)[Score](#)

NAND2Tetris / Hardware Simulator / Not.hdl

HDL ☒ --LBSt Project 1 Not **Chip Not** Eval Reset Clock: 0

```
1 // HDL (e.g., Hack HDL / Nand2Octet style)
2 CHIP Not {
3     IN in;
4     OUT out;
5
6     PARTS:
7         Nand(a=in, b=in, out=out);
8 }
```

Input pins
in 1

Output pins
out 0

Test Steps: 5 Outputs: 3 Slow Fast

Test Script Compare File Output File Diff Table

```
9
10 set in 0,
11 eval,
12 output;
13
14 set in 1,
15 eval,
16 output;
```

NAND2Tetris / Hardware Simulator / Not.hdl

HDL ☒ --LBSt Project 1 Not **Chip Not** Eval Reset Clock: 0

```
1 // HDL (e.g., Hack HDL / Nand2Octet style)
2 CHIP Not {
3     IN in;
4     OUT out;
5
6     PARTS:
7         Nand(a=in, b=in, out=out);
8 }
```

Input pins
in 0

Output pins
out 0

Test Steps: 5 Outputs: 3 Slow Fast

Test Script Compare File Output File Diff Table

```
9
10 set in 0,
11 eval,
12 output;
13
14 set in 1,
15 eval,
16 output;
```

ND Gate using NAND Gate

An AND gate outputs 1 only when both inputs are 1. Taking the output of a standard NAND gate and feeding it into a NAND-configured inverter (NOT gate) double-negates the operation back to AND.

CHIP And {

IN a, b;

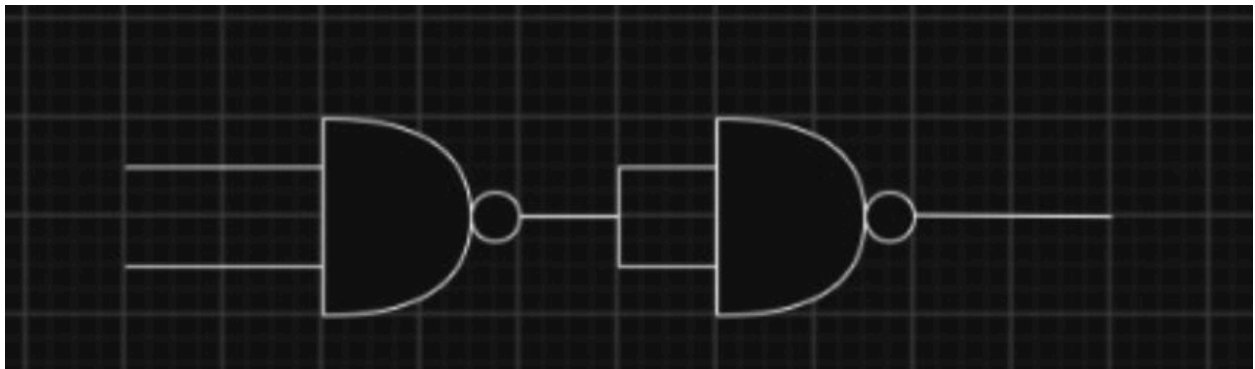
OUT out;

PARTS:

Nand(a=a, b=b, out=nandOut);

Nand(a=nandOut, b=nandOut, out=out);

}



HDL ConceptsHDL SyntaxInterfaceSimulatorTest ScriptsQuizScore

NAND2Tetris / Hardware Simulator / And.hdl

HDL --LBStProject 1And

1 CHIP And {
2 ... IN a, b;
3 ... OUT out;
4
5 ... PARTS:
6 ... Nand(a=a, b=b, out=nandOut);
7 ... Nand(a=nandOut, b=nandOut, out=out);
8 }

Chip And
EvalResetClock: 0

Input pins
a 1
b 0
Output pins
out 0
Internal pins
nandOut 1

TestSteps: 5 Outputs: 0SlowFast

Test ScriptCompare FileOutput FileDiff Table
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;

NAND2Tetris / Hardware Simulator / And.hdl

HDL --LBStProject 1And

1 CHIP And {
2 ... IN a, b;
3 ... OUT out;
4
5 ... PARTS:
6 ... Nand(a=a, b=b, out=nandOut);
7 ... Nand(a=nandOut, b=nandOut, out=out);
8 }

Chip And
EvalResetClock: 0

Input pins
a 1
b 1
Output pins
out 0
Internal pins
nandOut 1

TestSteps: 5 Outputs: 0SlowFast

Test ScriptCompare FileOutput FileDiff Table
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;

HDL ConceptsHDL SyntaxInterfaceSimulatorTest ScriptsQuizScore

NAND2Tetris / Hardware Simulator / And.hdl

HDL --LBStProject 1And

1 CHIP And {
2 IN a, b;
3 OUT out;
4
5 PARTS:
6 Nand(a=a, b=b, out=nandOut);
7 Nand(a=nandOut, b=nandOut, out=out);
8 }

Chip And
Input pins
a 0
b 0
Output pins
out 0
Internal pins
nandOut 1

TestSteps: 5 Outputs: 0SlowFast

Test ScriptCompare FileOutput FileDiff Table
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;

NAND2Tetris / Hardware Simulator / And.hdl

HDL --LBStProject 1And

1 CHIP And {
2 IN a, b;
3 OUT out;
4
5 PARTS:
6 Nand(a=a, b=b, out=nandOut);
7 Nand(a=nandOut, b=nandOut, out=out);
8 }

Chip And
Input pins
a 1
b 0
Output pins
out 0
Internal pins
nandOut 1

TestSteps: 5 Outputs: 0SlowFast

Test ScriptCompare FileOutput FileDiff Table
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;

NAND2Tetris / Hardware Simulator / And.hdl

HDL ☐ --LBSt Project 1 And Chip And Eval Reset Clock: 0

```
1 CHIP And {
2   ... IN a, b;
3   ... OUT out;
4
5   ... PARTS:
6   ... Nand(a=a, b=b, out=nandOut);
7   ... Nand(a=nandOut, b=nandOut, out=out);
8 }
```

Input pins

a 1

b 1

Output pins

out 0

Internal pins

nandOut 1

Test Steps: 5 Outputs: 0 Slow Fast

Test Script Compare File Output File Diff Table

```
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
```

[HDL Concepts](#) [HDL Syntax](#) [Interface](#) [Simulator](#) [Test Scripts](#) [Quiz](#) [Score](#)

NAND2Tetris / Hardware Simulator / And.hdl

HDL ☐ --LBSt Project 1 And Chip And Eval Reset Clock: 0

```
1 CHIP And {
2   ... IN a, b;
3   ... OUT out;
4
5   ... PARTS:
6   ... Nand(a=a, b=b, out=nandOut);
7   ... Nand(a=nandOut, b=nandOut, out=out);
8 }
```

Input pins

a 0

b 1

Output pins

out 0

Internal pins

nandOut 1

Test Steps: 5 Outputs: 0 Slow Fast

Test Script Compare File Output File Diff Table

```
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
```

OR Gate using NAND Gate

By applying **De Morgan's Laws**, an OR gate can be constructed by inverting both inputs using NAND-based NOT gates and then feeding those outputs into a final NAND gate.

CHIP Or {

IN a, b;

OUT out;

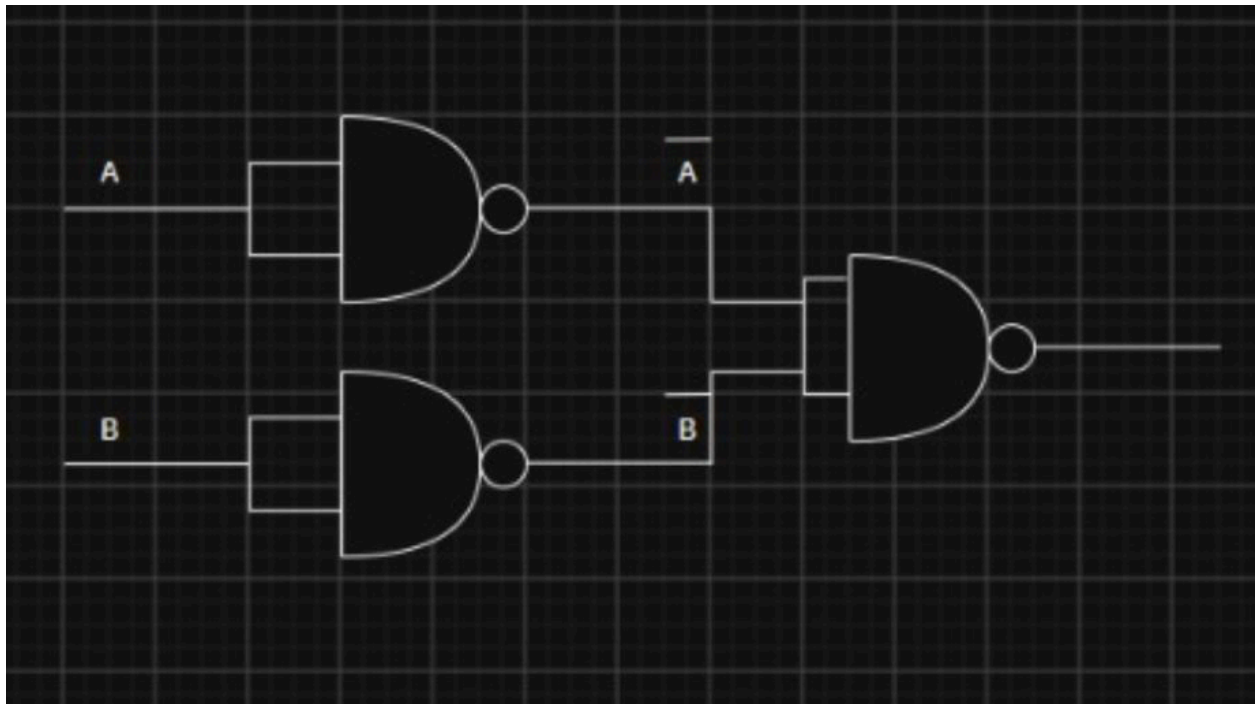
PARTS:

Nand(a=a, b=a, out=notA);

Nand(a=b, b=b, out=notB);

Nand(a=notA, b=notB, out=out);

}



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HDL

--LBSt

Project 1

Or

Chip Or

Eval

Reset

Clock: 0

1 CHIP Or {

2 IN a, b;

3 OUT out;

4

5 PARTS:

6 Nand(a=a, b=a, out=notA);

7 Nand(a=b, b=b, out=notB);

8 Nand(a=notA, b=notB, out=out);

9 }

Input pins

a 0

b 0

Output pins

out 0

Internal pins

notA 1

notB 1

Test

Steps: 5

Outputs: 0

Slow

Fast

Test Script

Compare File

Output File

Diff Table

21 set b 0,

22 eval,

23 output;

24

25 set a 1,

26 set b 1,

27 eval,

28 output;

NAND2Tetris / Hardware Simulator / Or.hdl

HDL ☒ --LBSt Project 1 Or   Chip Or Eval Reset Clock: 0

```
1 CHIP Or {
2   IN a, b;
3   OUT out;
4
5   PARTS:
6     Nand(a=a, b=a, out=notA);
7     Nand(a=b, b=b, out=notB);
8     Nand(a=notA, b=notB, out=out);
9 }
```

Input pins
a 1
b 1

Output pins
out 0

Internal pins
notA 1
notB 1

Test    Steps: 5 Outputs: 0 Slow Fast

Test Script Compare File Output File Diff Table

```
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
```

NAND2Tetris / Hardware Simulator / Or.hdl

HDL ☒ --LBSt Project 1 Or   Chip Or Eval Reset Clock: 0

```
1 CHIP Or {
2   IN a, b;
3   OUT out;
4
5   PARTS:
6     Nand(a=a, b=a, out=notA);
7     Nand(a=b, b=b, out=notB);
8     Nand(a=notA, b=notB, out=out);
9 }
```

Input pins
a 1
b 0

Output pins
out 0

Internal pins
notA 1
notB 1

Test    Steps: 5 Outputs: 0 Slow Fast

Test Script Compare File Output File Diff Table

```
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
```

NAND2Tetris / Hardware Simulator / Or.hdl

HDL

--LBSt

Project 1

Or

Chip Or

Eval

Reset

Clock: 0

```
1 CHIP Or {
2   IN a, b;
3   OUT out;
4
5   PARTS:
6     Nand(a=a, b=a, out=notA);
7     Nand(a=b, b=b, out=notB);
8     Nand(a=notA, b=notB, out=out);
9 }
```

Input pins

a 1

b 0

Output pins

out 0

Internal pins

notA 1

notB 1

Test

Steps: 5

Outputs: 0

Slow

Fast

Test Script

Compare File

Output File

Diff Table

```
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
```